in Haryana, Gujarat, Rajasthan and Uttar Pradesh published in the journal *Waterbirds*, the researchers, among them Program SarusScape director K.S. Gopi Sundar, observed unseasonal nesting among Sarus cranes (*Antigone antigone*) due to abnormal monsoonal conditions or when they found themselves in areas with artificial water resources.

Deforestation and the rise of monocultures are other prime reasons for birds losing out in the natural order of things. Monoculture plantations such as coffee, tea, rubber, teak or oil palm maintain single-tree species as they are easier to harvest. This leads to a loss of larger trees and tree cover, affecting birds that prefer higher or thinner branches. Studies in the dry deciduous and tropical thorn forests of Rajasthan as well as Himalayan oak forests have found that reduced canopy cover, undergrowth and tree height have adversely impacted bird diversity.

A RANGE OF STUDIES BEARS this out. Oil palm, for example, supports only 14 per cent of bird species that can be seen in rainforests next to them in Mizoram, according to a 2016 study in the journal *The Condor*. Only 50 per cent of woodpeckers in Uttarakhand's old-growth sal forests can survive in teak plantations that are taking over the state, as documented in a 2011 study in nature journal *Biological Conservation*. About 25 per cent of bird species are lost in tea plantations that come up next to rainforests. These monoculture plantations are expanding, going by a 2021 study in the publication *Current Science*. The area under oil palm plantations, for instance, has expanded 30-fold from 1991 to 2015, reports a 2021 study in the journal *Nature Food*, and most of this expansion is taking place in the tropical rainforest areas of the country.

SHRINKING WETLANDS

Waterbirds face as much threat as their forest cousins due to loss of habitat. India currently has 201,503 wetlands and most of them are under stress courtesy urbanisation. Riverine ground-nesting birds have declined by 50-80 per cent over the long term, according to the SoIB report, and some continue to decline even now. Crucial habitats like the Pulicat Lake on the Andhra Pradesh-Tamil Nadu border, the Gulfs of Khambhat and Kachchh in Gujarat, the Thane creek and adjoining mudflats in Mumbai and the Pallikaranai Marsh in Chennai are under threat of encroachment. The disruption in water levels on account of irrigation schemes, sand mining, climate change and pollution impacts the breeding of riverine sandbar-nesting birds. Trampling by grazing livestock that wander into islets when water levels recede leads to the destruction of 9-20 per cent of the eggs. Predation by dogs and jackals, who meander into nesting islands, also results in a loss of 32-35 per cent of the eggs, as noted by a 2020 study of the Bombay Natural History Society (BNHS).

Birds that live in open habitats are no safer. The decline of the Great Indian bustard (*Ardeotis nigriceps*) is a well-documented one. Critically endangered, only some 150 individuals are left in the wild. Several other species are on the decline. The Great Grey Shrike (*Lanius excubitor*), for instance, has exhibited a long-term decline of over 80 per cent. Experts attribute this to the invasion of open habitats by drought-tolerant woody plants as well as wind turbines and power lines. *Prosopis juliflora*, a prolific thorny invasive plant from South America known



ANDAMAN TEAL
Anas albogularis
Habitat: Wetland birds found in the Andaman & Nicobar Islands
Reason for decline: Hunting, reclamation of wetlands

HIGH

DECLINING DUCKS

TREND: India hosts eight resident and 35 migratory species of the bird in a range of habitats, including inland lakes and tanks, submerged paddyfields, rivers, forest pools and coastal lagoons. Both kinds are experiencing long-term and continued decline. The Common pochard is declining by 2.5 per cent annually; at Chilika Lake, the counts went from 90,000 in 2001 to 2,000 by 2014. The Andaman teal has a population under 1,000; Baer's pochard has a global population of less than 1,700 and very few migrate to India now. The Tufted duck has declined by nearly 80 per cent between 2014 and 2022.

WHY: Though there are no studies to pinpoint exactly why the duck population is threatened, hunting, loss of wetlands and avian disease could be among some of the reasons.